

PR TEST PROCEDURE



VERSION CONTROL		
VERSION	DATE	DESCRIPTION
0A	05/05/2026	Initial version

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1 PURPOSE

This document aims to define a procedure for measuring and analyzing the energy production of a specific photovoltaic system relative to expected electrical energy production for the same system from actual weather conditions, quantifying the energy performance index, as defined in IEC 61724-1.

The following procedures have been defined considering the provisions described in “ANNEX F5 – PR Methodology” of the contract.

2 PROJECT DATA

The project consists of a photovoltaic plant connected to the medium and high voltage electrical grid, to be installed on agricultural land with structures fixed into the ground and designed to optimize the capture of available solar energy.

Table 1. Project data.

Project Data	
Installation location	Fraga (Huesca)
Plant name	CF Fraga I
Peak power (MWp)	40,048 MWp
Nominal power (MWn)	33,00 MWn @ 40°C
Type of support structures	Single-axis trackers

The coordinates of the projects are:

Coordinates (UTM ETRS89 H31 N)				
PV Plant		Access to plant		Fenced area (m ²)
X	Y	X	Y	
270.820 E	4.598.304 N	271.283	4.598.271	536.615

Table 1. Coordinates.

The placement of the plants can be seen in the following image:

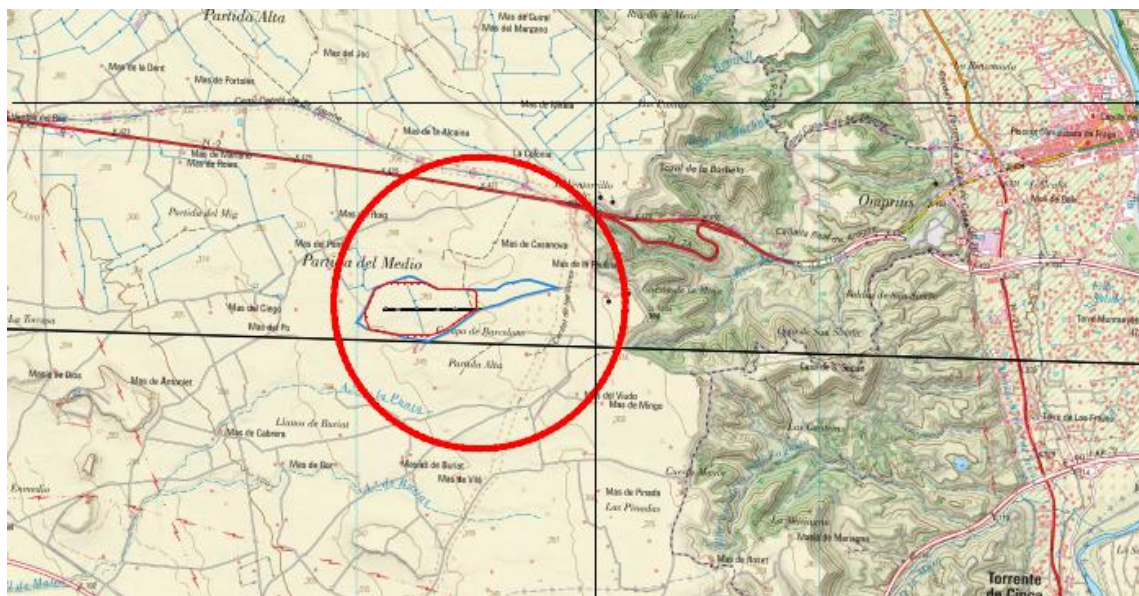


Imagen 1. PV plant's location.

3 CALCULATION METHOD

As a basis for this evaluation, expectations of energy production are developed using a model of the PV system under test that will serve as the guarantee or basis for the

evaluation and is agreed upon by all stakeholders of the project. The model is developed using the software PVSyst in its latest version. It will include real data of ambient temperature, albedo, diffuse and horizontal irradiation measured by the weather sensors installed within the plant, and must include effects of shading and variable efficiency of the array.

The energy production is characterized specifically for times when the system is operating (available); times when the system is not operating (unavailable) are quantified as part of an availability metric.

"EPI" Energy Production Index, as defined IEC 61724-3, it is the ratio between the Measured Energy and the Expected Energy as below:

$$EPI = \frac{\text{Measured Energy}}{\text{Expected Energy}} \quad (\text{Eq. 1})$$

In which,

"Expected Energy" is the energy output of the facility defined by the simulation model selected, considering the input of real weather conditions measured during the testing period, as defined in section 6.2. "Calculation and documentation of predicted energy and the method that will be used to calculate the expected energy" of the IEC 61724-3 standards.

"Measured Energy" is the Expected Energy measured at the meter during the test.

The duration of the Performance Test must be ten (10) consecutive full valid days. Should there be a system event or requirement, including among others weather conditions different than the minimum requirements established in this document, during the 10 days of the test, that register data will be discarded and the test will be extended. In this sense, the testing period will not be extended for more than 20 days in total. If such case happens, the performance test shall be restarted.

4 PREDICTED ENERGY AND MODEL

The first step in the process, typically, is to predict the performance of the PV system based on historical weather data using a model that has been agreed to by the stakeholders. The model is defined in terms of the model inputs, calculation process, and how the measured meteorological data will be input into the model. Although the final comparison of expected and measured energy does not use the predicted energy directly, the predicted energy is usually required for project planning. The model assumes 100% availability. The model agreed upon is defined below.

Table 2. Predicted model. According to the point 8.1 of Annex 3.

Meteorological database	Predicted values	Expected values
Meteorological database	SolarGIS TMY	Data measured on-site
System	Predicted values	Expected values
Module file 630W	JAM72D42-630_LB.pan	JAM72D42-630_LB.pan
Module file 635W	JAM72D42-635_LB.pan	JAM72D42-635_LB.pan
Inverter file	SUN2000-330KTL-H1-V0.2.ond	SUN2000-330KTL-H1-V0.2.ond
Modules per string	28	28
Number of modules	63.168	63.168
DC power	40,0484 MWp	40,0484 MWp
Number of inverters	110	110
AC power	33,00 MWac@30°C	33,00 MWac@30°C
Bifacial system definition		

Bifacial model	Unlimited trackers 2D-model	Unlimited trackers 2D-model
Ground albedo	0,24 (SolarGIS Prospect)	TBD
Structure shading factor	5%	9,90%
Mismatch loss factor	5,0%	5,4%
Bifaciality Factor	78%	78%
Unlimited trackers 2D model		
Axis azimuth	0°	0°
Pitch	6,00 m	6,00 m
Shed total width	2,44 m	2,47 m
Height above ground	1,47 m	1,47 m
Detailed losses	Predicted values	Expected values
Thermal parameters		
Constant Loss Factor U_c	29,00 W/m ² K	29,00 W/m ² K
Wind Loss Factor U_v	0,00 W/m ² K/m/s	0,00 W/m ² K/m/s
Ohmic losses		
DC circuit: Loss fraction at STC	0,60%	0,60% ¹
DC circuit: Serie Diode Loss	0	0
AC LV circuit: Loss fraction at STC	1,50%	0,46% ²
AC MV circuit: Loss fraction at STC	0,50%	0,25% ³
MV external transformer: Iron Loss	0,10%	0,068% ⁴
MV external transformer: Copper (resistive) Loss	1,00%	0,62%
HV external transformer: Iron Loss	0,10%	(0,0165 MVA) 0,05%
HV external transformer: Copper (resistive) Loss	1,00%	(0,1313 MVA) 0,38%
Modules quality		
Efficiency loss	0,00%	0,00%
LID loss factor	0,60%	0,60%
Module mismatch losses (Power losses at MPP)	1,00%	1,00%
Strings voltage mismatch (Power Loss at MPP)	0,10%	0,10%
Soling losses		
Yearly loss factor	1,50%	1,50%
IAM losses		

¹ According to "FRA.FV.MEM.ELE.DC.004 Memoria de calculo cable DC".

² According to "FRA.FV.MEM.ELE.DC.006 Memoria de calculo cable AC".

³ According to "FRA.FV.MEM.ELE.MT.009 Memoria de calculo cable MT".

⁴ All the transformer losses are according to FAT tests.

Uses definition of the PV Module	CHECK "Uses definition of the PV Module"	CHECK "Uses definition of the PV Module"
Auxiliaries		
Proportional to output power	4 W/kW	4 W/kW
Night auxiliares consumption	0 kW	0 kW
Aging		
Do not include in PVsyst. It is applied to the results of the software.	0%	0%
Unavailability		
Do not include in PVsyst. It is applied to the results of the software.	0%	0%
Spectral correction		
Do not include in PVSyst	DO NOT CHECK	DO NOT CHECK
Horizon	Predicted values	Expected values
Horizon profile	Free horizon	Free horizon
Near Shadings	Predicted values	Expected values
Use in simulation	CHECK "According to module strings"	CHECK "According to module strings"
Calculation mode	CHECK "Slow (simul.)"	CHECK "Slow (simul.)"
Fraction for electrical effect	100%	100%
Construction / Perspective	Backtracking active Get a layout like the shape of the field Tracker inclination according to manufacturer	Backtracking active Get a layout like the shape of the field Tracker inclination according to manufacturer
Energy management	Predicted values	Expected values
Inverter temperature		
Inverter temperature	CHECK External ambient temperature (outdoor installation)	CHECK External ambient temperature (outdoor installation)
Power factor		
Power factor	1,00	1,00
Power factor mode	CHECK "Use a Power Factor for grid injection"	CHECK "Use a Power Factor for grid injection"
Grid limitation		
Power limitation	29,04 Mwac	29,04 Mwac ⁵

⁵ According to client's information

Grid power limitation	CHECK "Uses grid power limitation" CHECK "Account as separate loss" CHECK "Limit applied at the injection point" Indicate the POI power in "Grid Power Limitation"	CHECK "Uses grid power limitation" CHECK "Account as separate loss" CHECK "Limit applied at the injection point" Indicate the POI power in "Grid Power Limitation"
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5 EXPECTED ENERGY

5.1 Data acquisition

5.1.1 Instrumentation

The information regarding the meteorological conditions on site will be measured by the weather sensors installed on the meteorological stations and on the trackers relative trackers. Additional information is contained in Appendix I.

Meteo Station 1 "P05.MET.01-01"

Table 3. Meteo sensors of station type 1.

Parameter	Type of sensor	Number of sensors
Horizontal global irradiance	NRG Systems R2-D SMP10-GHI	2
Plane-of-array irradiance	NRG Systems R2-D SMP10-GTI (P01.TR024 Drive row & Slave Row)	2
Diffuse horizontal irradiance	Delta-T SPN2	1
Ambient temperature	Thies Clime Sensor US Multimet	1
Panel Temperature	NRG Systems PVT1 PT100 BOM-Temperature	6
Reflected Horizontal Irradiation	NRG Systems R2-D SMP10-V	1
Soiling	2 Cells of irradiance	1

Meteo Station 2 "P05.MET.02-01"

Table 3. Meteo sensors of station type 2.

Parameter	Type of sensor	Number of sensors
Horizontal global irradiance	NRG Systems R2-D SMP10-GHI	2
Plane-of-array irradiance	NRG Systems R2-D SMP10-GTI (P05.TR057 Drive row & Slave Row)	2
Diffuse horizontal irradiance	Delta-T SPN2	1
Panel Temperature	PT100 Temp Panel	6
Wind speed	Anemometer & Wind vane NRG Systems #40 + 200M	1

Parameter	Type of sensor	Number of sensors
Reflected Horizontal Irradiation	NRG Systems R2-D SMP10-V	2
Soiling	2 Cells of irradiance	1

The meteorological station indicated in the table above, is positioned in the following coordinates:

Table 4. Meteorological station positions.

Coordinates (UTM ETRS89 H31 N)			
Meteo station 1 (P05.MET.01-01)		Meteo station 2 (P05.MET.02-01)	
X	Y	X	Y
270.283 E	4.598.277 N	270.882 E	4.598.263 N

The albedometer is comprised of two pyranometers: One facing upward, which measures the incoming global solar radiation from the sky (both direct sunlight and diffuse light), and one facing downward, which measures the reflected solar radiation from the surface below. Given that, the albedo is finally calculated according to the equation below:

$$ALB = \frac{REF}{GHI} \quad (\text{Eq. 2})$$

5.1.2 Contractual requirements

Sampling period is hereby defined as the interval in which the different variables (Measured Energy, DHI, GHI, Ambient Temperature) are recorded in the Facility for the purpose of calculating the Predicted Energy and the Expected Energy. This period is set to 15 min. For the avoidance of doubt, hourly values of Ambient Temperature, DHI and GHI will be calculated as an average of the values measured during the 15-min intervals within each hour. In the case of Measured Energy, hourly values will be calculated as the sum of the values measured during the 10 min intervals within each hour.

Irradiance will be measured as the average of all the horizontal pyranometers installed at the Facility. All pyranometers shall comply with ISO-9060 "Secondary Standard" certification. The pyranometers shall be cleaned every day during the ten (10) day test period (or extended period, if required).

5.1.3 Data format and quality

Processed data shall be prepared in an Excel spreadsheet with a frequency equal to the sampling period with the following columns:

- Time and date; Year/Month/day Hour: Minute
- GHI measured in all horizontal pyranometer.
- DHI measured in all horizontal diffusometers.
- Ambient temperature measured in all temperature sensors.
- Meter output measured by the meter in 110kV.

The spreadsheet will be readable (values and formulae) by the Owner for the purposes of checking.

The EPC Contractor will supply raw data before any manipulation and highlight any gaps in the data to the Owner and Owner's personnel.

The sampling period in which the average value is irrational or returns null values shall render the Project as unavailable in its entirety unless the lack of data or data rationality can be appropriately accommodated for. If possible, the lack of data will be accommodated using SolarGIS data series for the time step in which the average value is null, in agreement between the Owner and the EPC Contractor. If this time step is not valid due to not complying with any of the criteria described in this document, the data substitution will not be done.

5.2 Criteria for a successful test

The irradiance and ambient temperature used as input to the model should be the average of the available measurements, except where one measurement is determined to be erroneous, in which case the input to the model should be the average of the remaining measurements. In order to detect and remove outlier values, the Interquartile Range (IQR) Method was used. This is a robust method less affected by extreme values.

Outliers are typically defined as data points that comply with the following criteria:

$$\text{Lower Outlier} < Q1 - 1.5 * IQR \quad (\text{Eq. 3})$$

$$\text{Upper Outlier} > Q3 + 1.5 * IQR \quad (\text{Eq. 4})$$

In which,

Q1 is the first quartile

Q3 is the third quartile

IQR is the interquartile range

Once the outliers are identified, they are removed from the database and the average of the remaining sensors is calculated. This process is executed for both GHI and temperature.

5.2.1 Criterion 1

The average of all pyranometers measurements will be calculated during the test period, once removed all the outliers according to the previous point. For those pyranometers that disagree by greater than 5% with respect to the daily average of all pyranometers, the values will be discarded. In each sampling period, the GHI of the Facility will be calculated as the average of all the valid diffusometer.

5.2.2 Criterion 2

Those diffusometers that disagree by greater of 5% with respect to the daily average of all diffusometers installed will be considered not valid. Even within this 5%, each diffusometer having non-consistent data will be also considered as not valid. In each sampling period, the DHI of the Facility will be calculated as the average of all the valid diffusometer.

5.2.3 Criterion 3

Where the ambient temperature sensors disagree by greater than 10% and 5°C of the average over the sampling period the outlying results shall be disregarded.

5.2.4 Criterion 4

The sampling period each day will cease when no further energy from the inverters is being received, or the plane-of-array irradiance is below 100W/m². *On each day of the test, the test will be deemed to have commenced at this minimum threshold.*

5.2.5 Criterion 5

During the Performance Test period, there shall be at least 10 hours of plane-of-array irradiance of 600 W/m² measured in the period from the start of April to the end of September. If the tests are executed between the first day of October and the last day of March, there shall be at least 10 hours of plane-of-array irradiance of 400 W/m² measured within the entire Performance Test period.

5.2.6 Criterion 6

During the test, any limitation or unavailability not attributable to the Contractor should be removed from the calculations, as described below:

- Owner's Instructions
- Utility curtailment of the PV Plant
- Damages to the Project or Defects out of the Warranty
- Network outages
- Governmental Authority instructions
- Force Majeure
- Stow position

5.3 Expected production calculation

The simulation will be carried out on the latest version of PVsyst, since this most updated revision of the software ensures a more accurate estimation of the plant performance (8.1.1).

The model of the simulation will consider the as-built conditions of the layout, according to the "Table 2. Predicted model.

The 15-minute weather data, more specifically the GHI, DHI and Tamb, resulting from the filtering process described above will be inserted in PVSyst as a custom weather file.

6 MEASURED PRODUCTION ACQUISITION

The measured production will be extracted directly from the meter located at the high voltage side of the main transformer (110 kV).

7 TEST COMPLETION AND RESULTS

For the test period, expected and measured energy are determined by summing up the energy calculated in PVSyst and the measured AC energy over all eligible 15-minute periods. After the definition of the expected and the real production, it is possible to carry out the comparison of the results, in the period of reference.

$$\text{Measured EPI} \geq \text{Guaranteed EPI} \quad (\text{Eq. 5})$$

In which,

$$\text{Guaranteed EPI} = 98\% \times \text{Expected EPI} \quad (\text{Eq. 6})$$

7.1 Availability

The availability according to the methodology described in this document will be calculated for the peak power (kWp) not affected by an External Factor, defined below. For the sake of clarity, for each hour of the availability calculation, the peak power affected by an External Event will be excluded from the calculation.

The actual availability of the Project will be calculated as described below:

$$AVA = \frac{\sum_{G_h > 100 \text{ Wh/m}^2} AVA_h \cdot G_h}{\sum_{G_h > 100 \text{ Wh/m}^2} G_h} \quad (\text{Eq. 7})$$

Where:

- G_h Is the global radiation measured in the plane of the array [Wh/m²] during hour h. Only periods in which G_h is greater than 100 Wh/m² will be considered for the calculation.

G_h will be calculated as the average of the individual measurements of the pyranometers installed on the array plane, if:

- there are no measurement errors
- none of the values deviate from the hourly average by more than 5%.

Any measured data that meet one of the two mentioned conditions above will not be considered at that hour and G_h will be calculated from the remaining measurements.

- AVA_h is the availability of the Project within the considered hour h.

The availability of the project within the considered hour h is calculated as follows:

$$AVA_h = \frac{\sum_{i=1}^4 AVA_{zh} \cdot P_{pz}}{\sum_{i=1}^4 P_{pz}} \quad (\text{Eq. 8})$$

Where:

- P_{pz} is the peak power connected to the zone "Z" (kWp)
- AVA_{zh} is the availability of the zone "Z" within the hour "h". Calculated as follows:

$$AVA_{zh} = \frac{\sum_{i=1}^N AVA_{izh} \cdot P_{piz}}{\sum_{i=1}^N P_{piz}} \quad (\text{Eq. 9})$$

Where:

- P_{piz} is the peak power connected to the inverter "i" (kWp) of the zone "Z".
- AVA_{izh} is the availability of inverter "i" of the zone "Z" within the hour "h", equal to:
 - $AVA_{izh} = 1$ if equivalent production of inverter "i" of the zone "Z" within the hour "h" is above 0 and EP_{piz} is above $0.9EP_{maxzh}$

- $AVA_{izh} = EP_{piz} / EP_{maxzh}$ if equivalent production of inverter "i" of the zone "Z" within the hour "h" is above 0 and EP_{piz} is below $0.9EP_{maxzh}$

- N is the total number of inverters
- h is the number of hours in which the irradiation values are above 100 Wh/m^2

In order to calculate AVA_{izh} ,

- EP_{maxzh} is the maximum of the equivalent production values among the inverter "i" of each zone of the Project during hour "h". This value represents the equivalent production of the inverter with the highest performance during hour "h" for each of the zones. For the avoidance of doubt, equivalent production is understood as the ratio between the maximum production in kWh and the peak power in kWp.
- EP_{piz} is the inverter "i" of zone "Z" equivalent production within hour "h" calculated as follows:

$$EP_{izh} = \frac{P_{izh}}{P_{piz}} \quad (\text{Eq. 10})$$

In which,

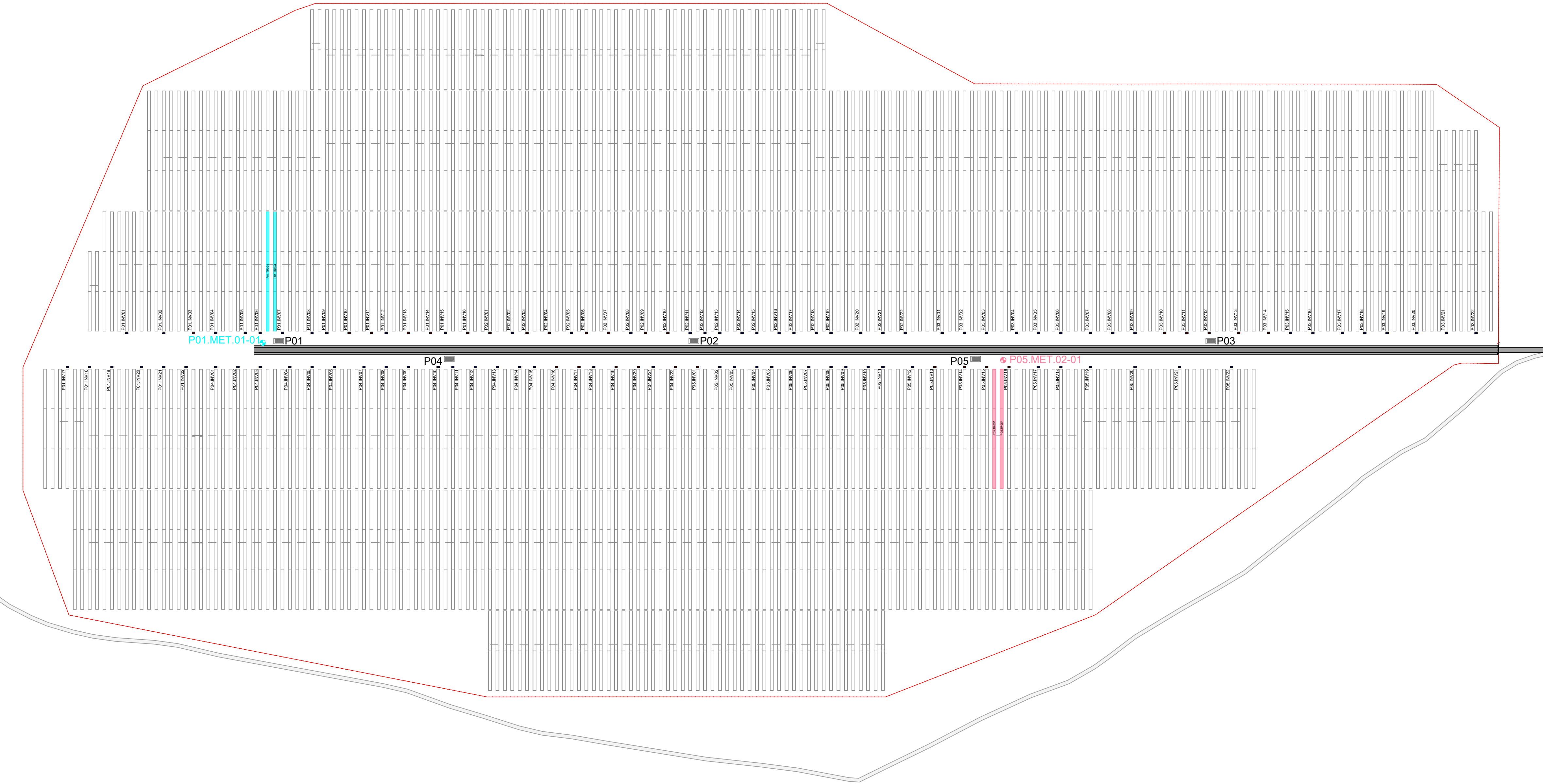
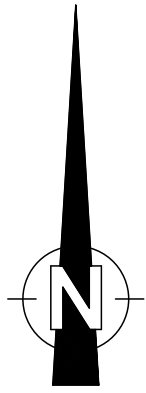
- P_{izh} is the Production (kWh) of inverter "i" of zone "Z" within the hour "h".
- P_{piz} is the peak power (kWp) connected to inverter "i" of zone "Z".

The system shall have a guaranteed availability of 100% during the Provisional Acceptance Test.

If any of the data indicated above is not available for more than 36 hours continuously for reasons not considered as an External Factor, the inverter for which the unavailability of data occurs will be considered unavailable ($AVA_{zh}=0$) during that entire period. In any case, if unavailable data could be obtained by other means, these may be used for the purpose of calculating availability.

In the event of missing data for periods shorter than 36 hours continuously, the inverter for which the missing data occurs will be excluded from the availability calculation, up to a maximum of 88 non-continuous hours per year. Should that event occur, liquidated damages of any kind related to this delay will not apply.

APPENDIX I - LOCATION AND DETAILS OF THE METEO STATIONS



LEYENDA AC

- VALLADO
- PUERTA DE ACCESO CORREDERA
- PUERTA DE ACCESO PEATONAL
- CAMINO INTERNO 4m ANCHO + 1m A CADA LADO PARA DRENAJE
- CAMINO DE ACCESO
- CENTRO DE TRANSFORMACIÓN
- SEGUIDOR 1V56
- SEGUIDOR 1V84
- EST. METEOROLÓGICA TIPO 1
- EST. METEOROLÓGICA TIPO 2

CUADRO DE COORDENADAS

EPSG:25831 - ETRS89 / UTM zona 31N

PUNTO	X	Y
P01.MET.01-01	270283.372	4598277.002
P05.MET.02-01	270882.715	4598263.022

CUADRO DE LOCALIZACIÓN

PUNTO	NÚMERO DE TRACKERS
P01.MET.01-01	P01.TR024
P05.MET.02-01	P05.TR057

DOCUMENTOS RELACIONADOS

- FRA.FV.DRW.COM.EM.007 DETALLE ESTACIÓN METEOROLÓGICA
- FRA.FV.DRW.COM.EM.008 ESQUEMA UNIFILAR ESTACIÓN METEOROLÓGICA
- FRA.FV.MEM.COM.EM.007 MEMORIA DE DISEÑO ESTACIÓN METEOROLÓGICA

NOMENCLATURAS

Pzz	Número de Subplanta	Pzz.INVyy	Número de inversor
			Número de Subplanta
Pzz-METyy-xx	Número de Orden		
	Número de Tipo de Estación Meteorológica		
	Número de Subplanta		

0 10 20 40 60 80 100 120 140 160m

Escala: 1/2000

NOTAS

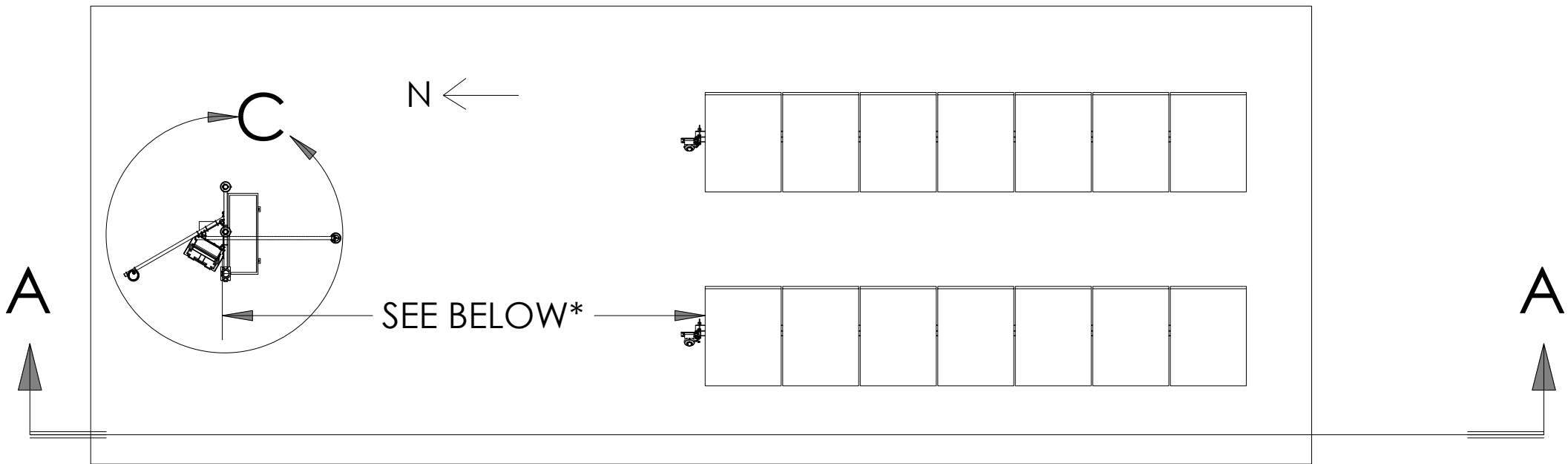
- Para ubicar la estación meteorológica, se deben tener en cuenta los siguientes parámetros:
- Medidas
- La ubicación debe ser tal que las medidas obtenidas sean representativas de todos el proyecto.
- Irradiación
- No puede haber elementos que hagan sombra sobre los elementos de la estación: piranómetros, células de radiación y panel solar.
 - Se debe evitar estar cerca de viales internos o carreteras, para que los sensores no acumulen suciedad, afectando las mediciones de radiación.
 - La estación no debe hacer sombra sobre los módulos.
- Viento
- Hay que tener en cuenta que la estación lleva un anemómetro y, si se instala cerca de un edificio, estos objetos podrían causar interferencias en las mediciones.
 - Siempre que sea posible, la torre debe ser ubicada de tal manera que, en la dirección predominante del viento, la estación esté libre de obstáculos. En caso de no ser posible, el anemómetro debe ser instalado a una altura de 2 veces la altura del obstáculo más cerca.
- Cableado
- Dependiendo del cable de alimentación también se debe tener en cuenta la distancia máxima.
 - La distancia máxima de los cables de las sondas es de 50 metros. O sea, no debe haber una distancia superior desde los paneles o estructura hasta la estación meteorológica superior a 40 metros.

PRODUCTO	DESCRIPCIÓN	CANTIDAD
ESTACIÓN METEOROLÓGICA TIPO 1	- 2 x Piranómetro POA SMP 10 o similar con 50 metros de cable y soporte. - 2 x Piranómetro GHI SMP 10 o similar con 1,0 metros de cable y soporte. - 1 x Piranómetro Secondary standard SMP 10 + kit RaZON + para radiación difusa. - 6 x Dispositivo PT100 para medir la temperatur del módulo con 50 metros de cable. - 1 x Sonda de temperatura ambiente, humedad y presión. - 1 x Anemómetro y veleta. - 1 x Pluviómetro. - 1 x higrómetro + barómetro - 1 x Albedómetro formado por dos piranómetros SMP 10 - 1 x Sensor de soiling basado en 2 células de irradiancia - 1 x Cuadro de datalogger. - 1 x Mástil y accesorios de montaje. - 1 x UPS + módulo fotovoltaico	1
ESTACIÓN METEOROLÓGICA TIPO 2	- 2 x Piranómetro POA SMP 10 o similar con 50 metros de cable y soporte. - 2 x Piranómetro GHI SMP 10 o similar con 1,0 metros de cable y soporte. - 1 x Piranómetro Secondary standard SMP 10 + kit RaZON + par radiación difusa. - 6 x Dispositivo PT100 para medir la temperatur del módulo con 50 metros de cable. - 1 x Anemómetro y veleta. - 1 x Albedómetro formado por dos piranómetros SMP 10 - 1 x Sensor de soiling basado en 2 células de irradiancia - 1 x Cuadro de datalogger. - 1 x Mástil y accesorios de montaje. - 1 x UPS + módulo fotovoltaico	1

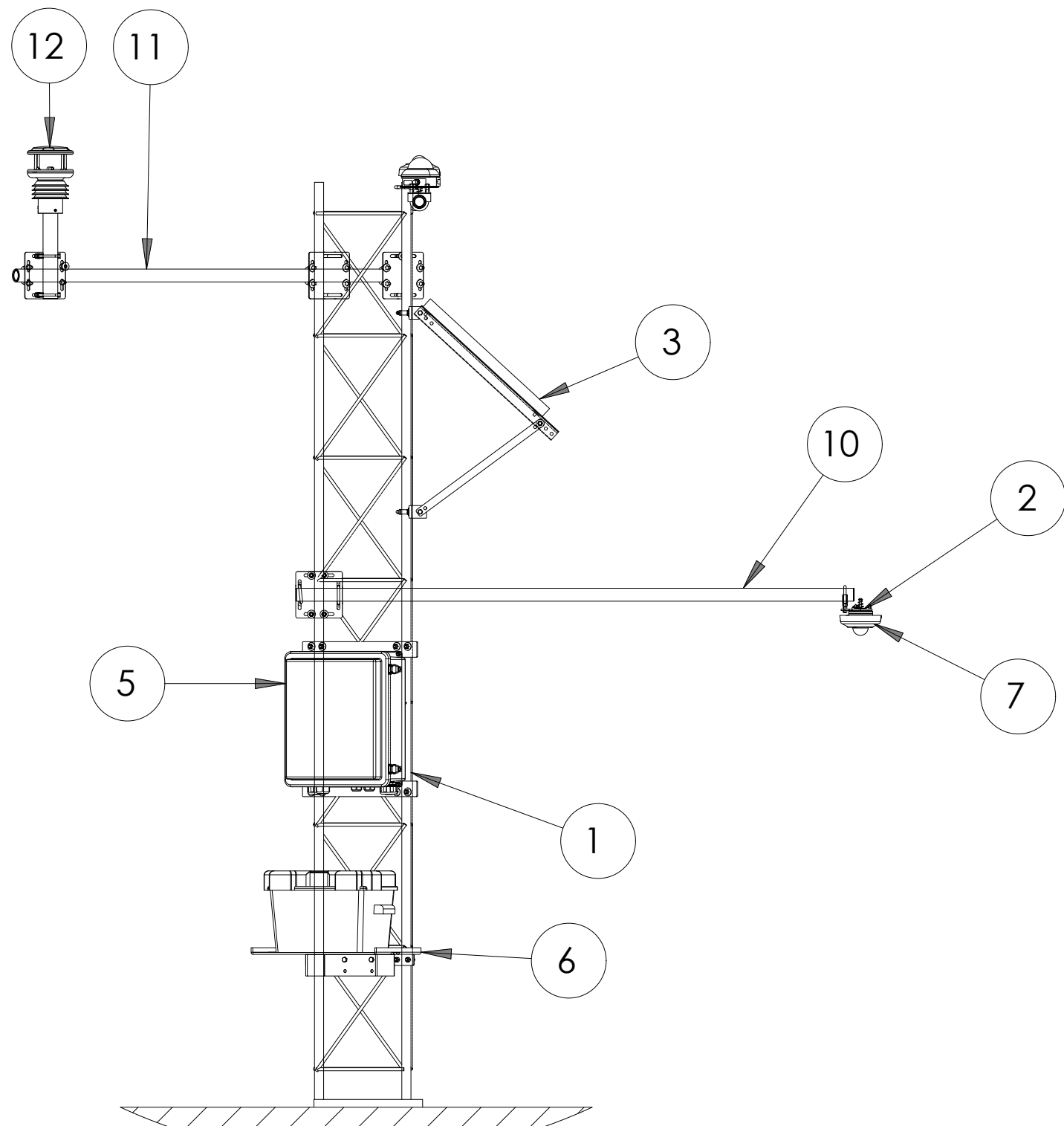
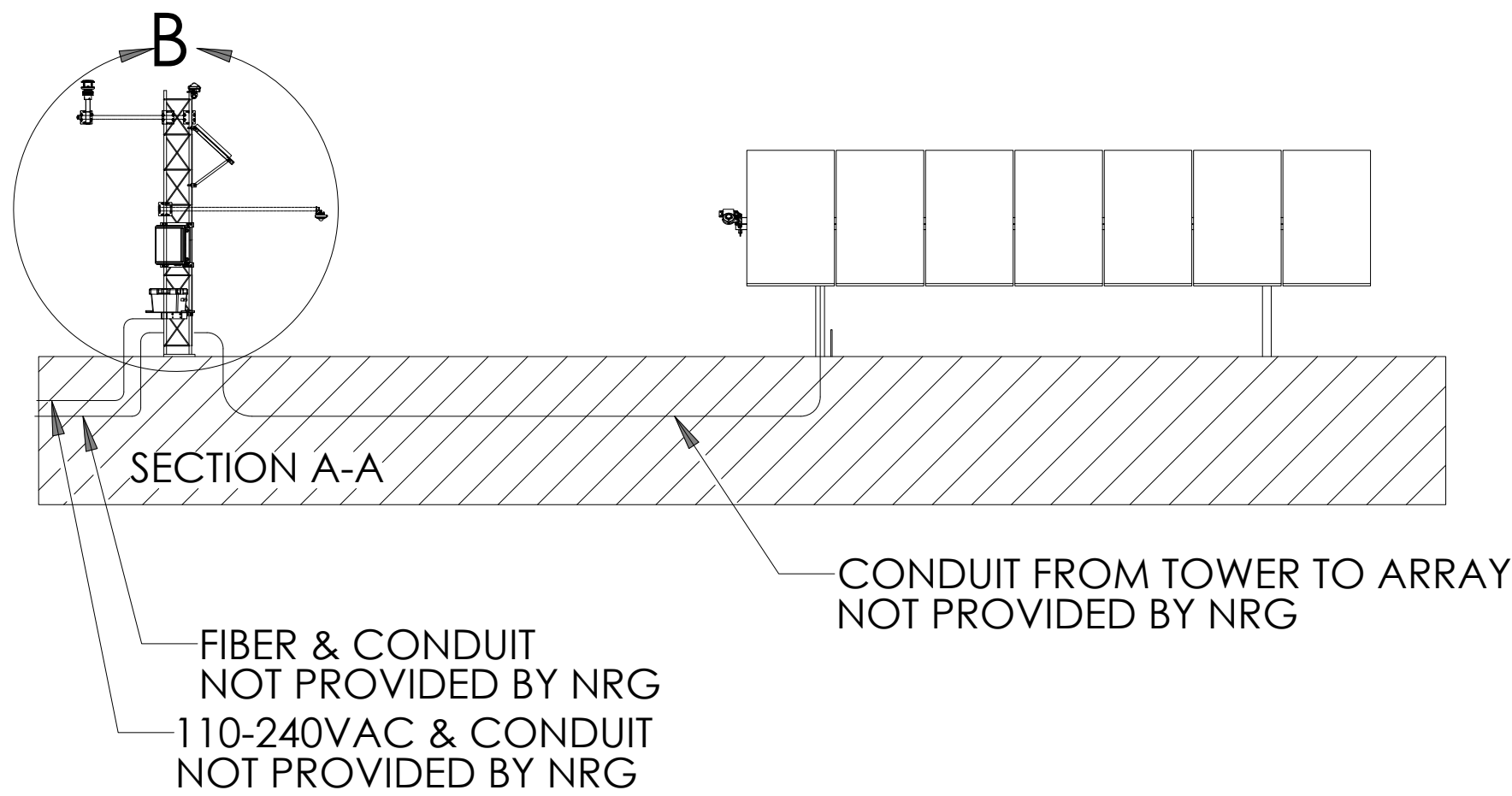
FORMATO	A1	1A	13/02/2026	VCR	PMN	PJL	PSS	As built
ESCALA	00	23/06/2025	VCR	MLP	PJL	PSS	Válido para construir	
1/2000	0A	06/05/2025	VCR	JAN	PJL	PSS	Emisión Inicial	
Rev.	Fecha	Diseñado	Dibujado	Revisado	Aprobado	Descripción		
PLANTA FOTOVOLTAICA SOLAR FRAGA, HUESCA								CODIGO: FRA.FV.DRW.COM.EM.006
UBICACIÓN DE ESTACIONES METEOROLÓGICAS								REVISION: 1A
								HOJA: 01

TOWER OVERVIEW
BOTH STATION TYPES

ARRAY SHOWN FOR ILLUSTRATIVE PURPOSES ONLY. PANELS & TRACKER NOT PROVIDED BY NRG. TOWER MAY BE LOCATED IN A DIFFERENT ORIENTATION RELATIVE TO ARRAY



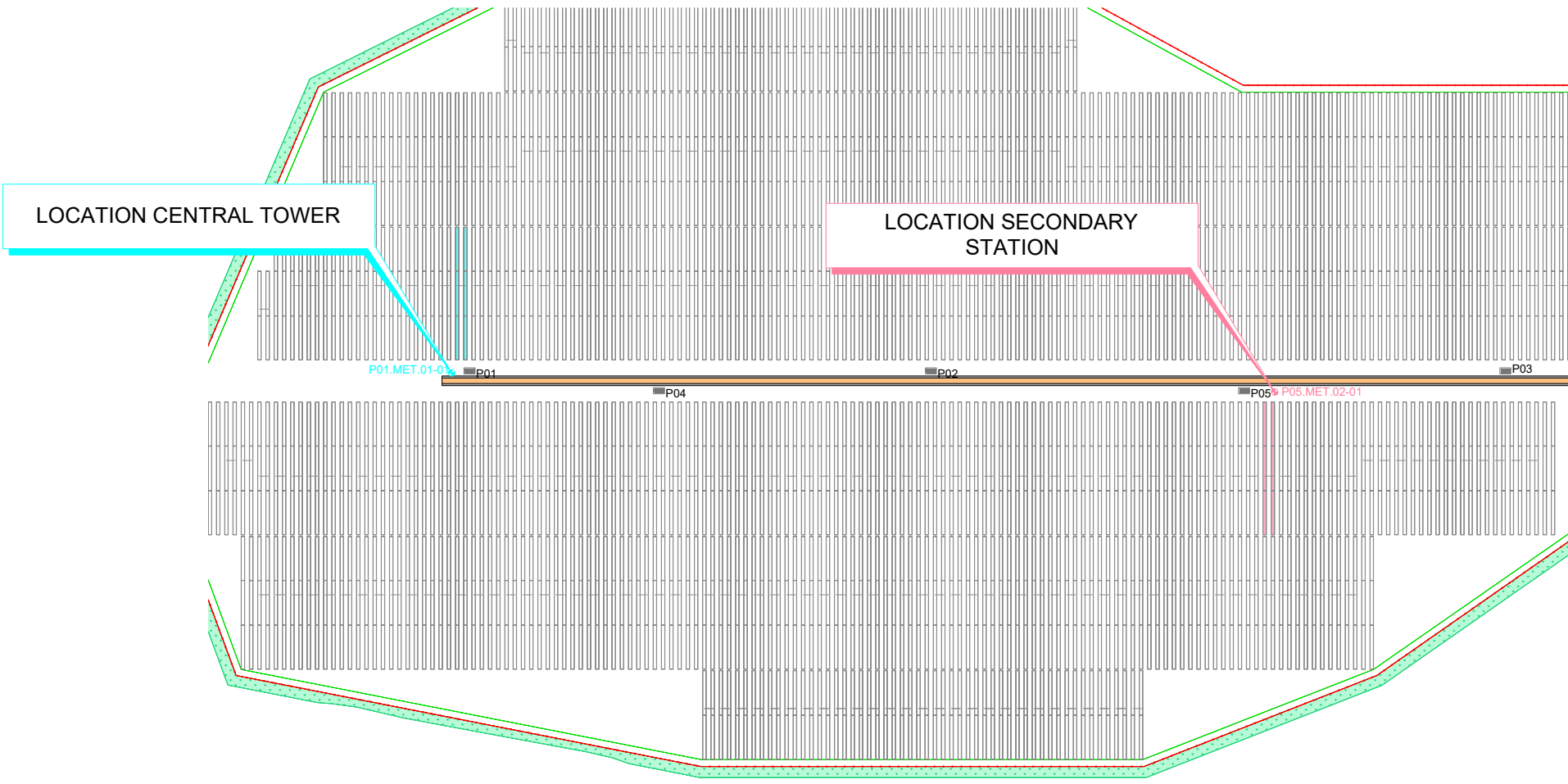
*THE LAYOUT OF THE TOWER (HEIGHT AND WIDTH) SHOULD BE CONSIDERED CAREFULLY BEFORE PLACEMENT IS DETERMINED. EACH SITE IS UNIQUE AND DETAILS SUCH AS SHADING (THE ARRAY SHOULD NEVER BE SHADED BY THE TOWER) AND CONDUIT RUNS FROM POWER/NETWORK CABLES TO THE ARRAY SHOULD BE REVIEWED. THESE DRAWINGS DO NOT TAKE EACH SITE LOCATION INTO CONSIDERATION.



DETAIL B

ELECTRONICS TO BE GROUNDED TO ISOLATED GROUND POINT. NRG RECOMMENDS USING STANDALONE GROUND ROD (PROVIDED BY SITE)

24" CONDUIT STUB-UPS RECOMMENDED WITH FLEX CONDUIT CONNECTION TO SHELTER BOX (NOT PROVIDED BY NRG)



Lookup ID	Description	Item Number	Qty. Complete Station	Qty Secondary Station
1	Lattice Tower, Lasser, 3m Flat top, C4	20219	1	1
2	NRG Pyranometer Pipe Mount Assembly	14357	3	3
3	RPS, One Panel, Lattice Tower	17029	1	1
4	Assembly, Lasser Lattice Tower, 4ft Boom	15946	1	1
5	Power Supply, Enclosure + LOGR Solar + Lattice Mounting Assembly	18324 + 9432 + 17170	1	1
6	Assembly, Lattice Tower Mounted Battery Bracket	18287	1	1
7	Sensor, Pyranometer, NRG R2-D, Digital	9462	5	5
8	Assembly, Pipe Mount, Pyranometer, Universal	14502	1	1
9	Delta-T Pyranometer, SPN1	10535	1	1
10	Assembly, RHI, Lattice Tower	16953	1	1
11	Assembly, Lattice Mounted Calypso	20136	1	1

Lookup ID	Description	Item Number	Qty. Complete Station	Qty Secondary Station
12	Thies Ciima MultiMet	19501	1	1
13	PV Sensor, Reference Cell, IMT	15496	2	2
14	Sensor, Temperature, PV Panel, 50 m Cable	9424	6	5
15	Array Assembly, NRG SP100, Surge Device	20304	2	2
16	Rod-Ground, 5/8" x 4', Copper Clad Steel	1533	2	2
17	IMT Digital Back of Module Temp Sensor	19086	0	1
18	#40 Anemometer	14881	0	1
19	200M Wind Vane	14861	0	1
20	Assembly, POA Ref Cell and Pyranometer	17830	2	2
21	Side Mount Z Boom, Lattice, 0.9m	16937	0	2

DOCUMENTOS RELACIONADOS

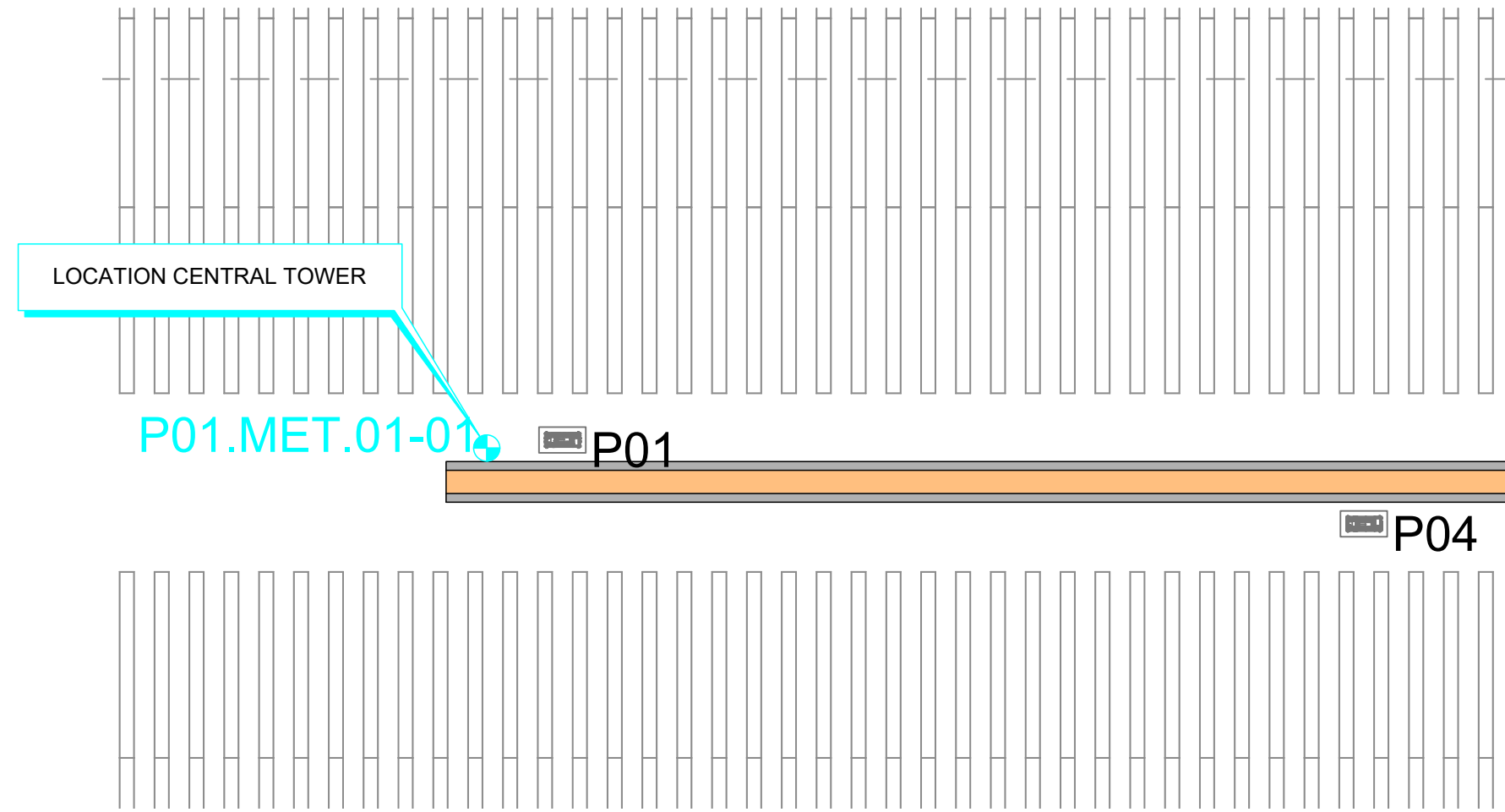
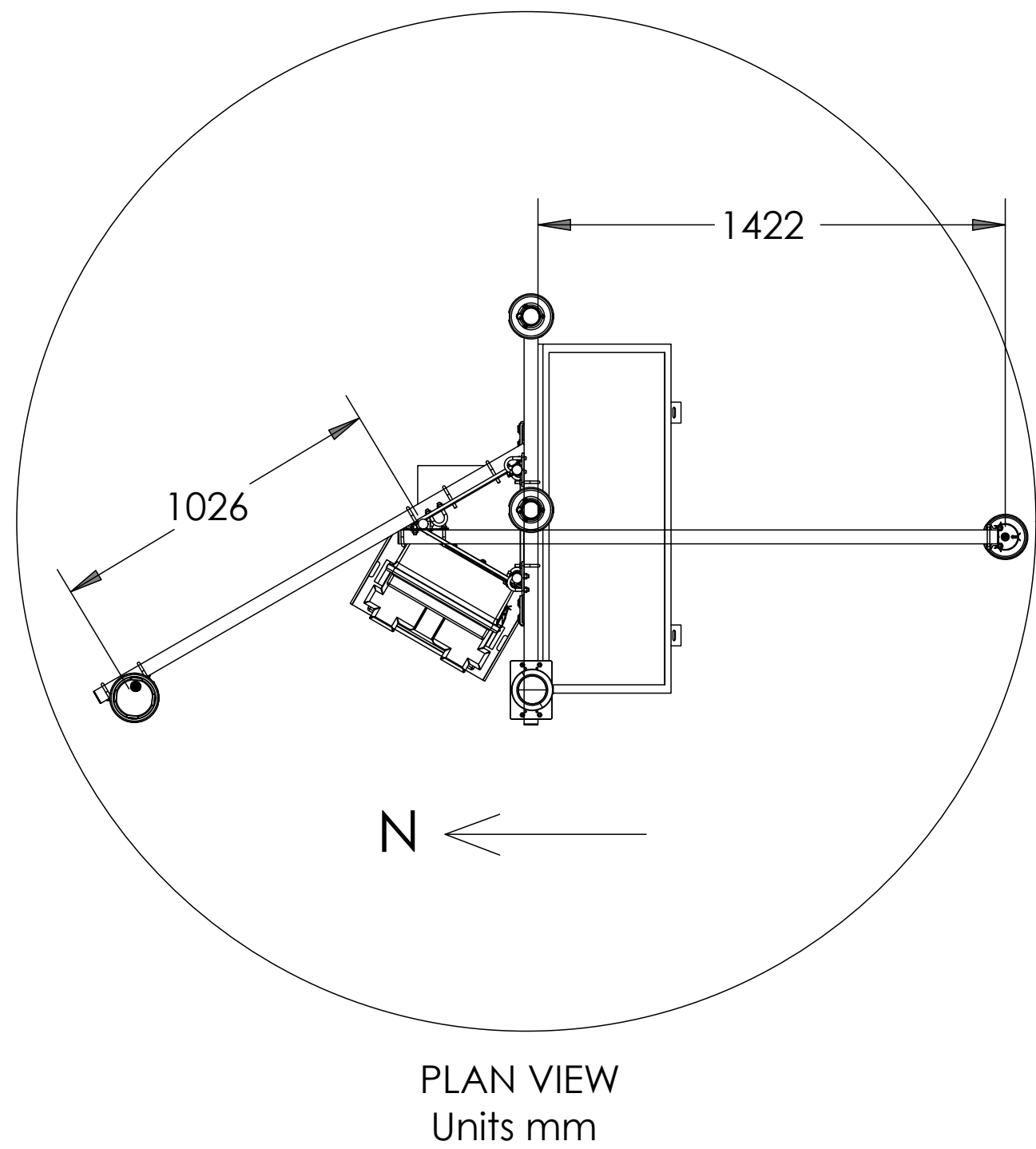
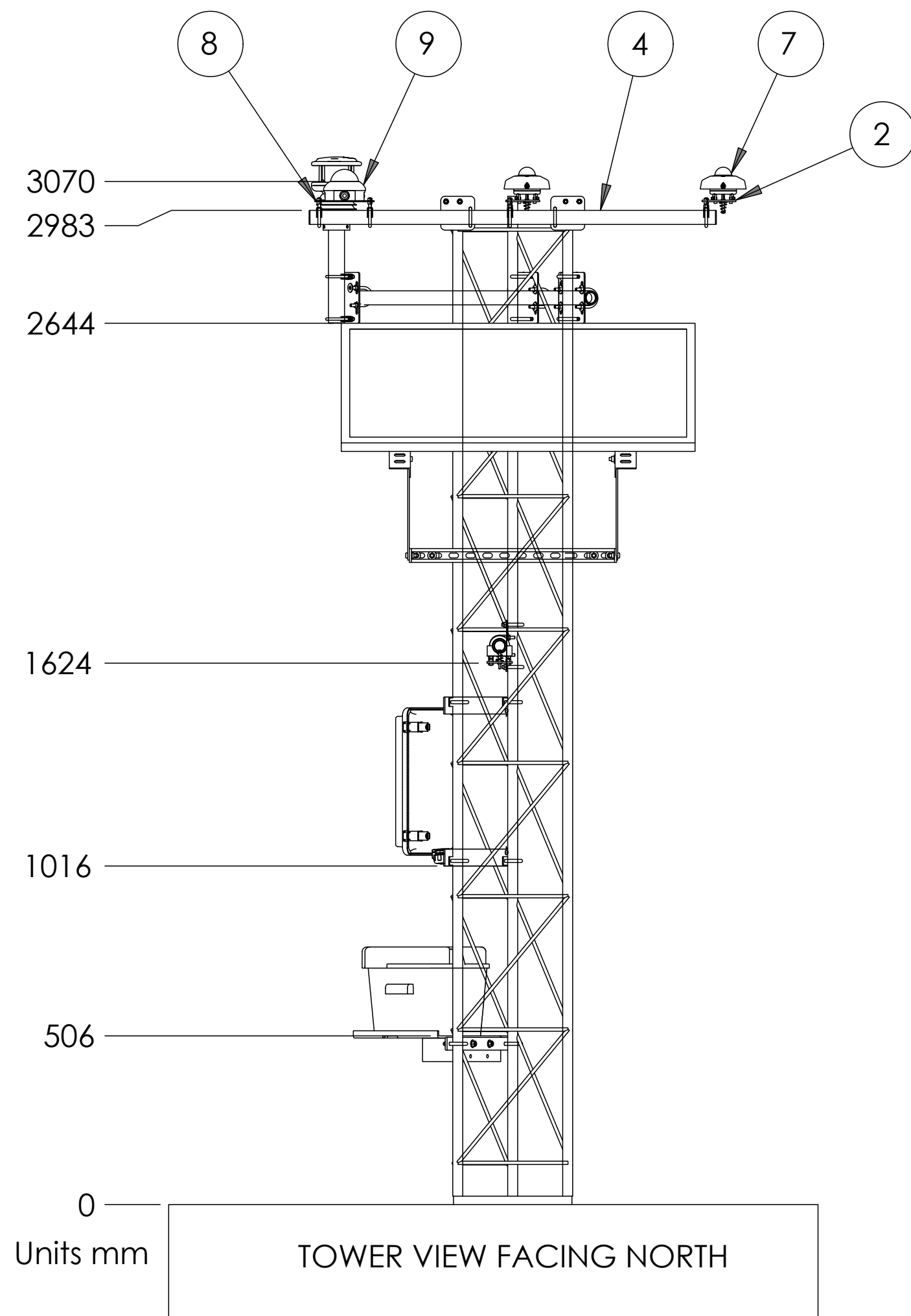
- FRA.FV.DRW.COM.EM.006 UBICACIÓN DE ESTACIONES METEOROLÓGICAS
- FRA.FV.DRW.COM.EM.008 ESQUEMA UNIFILAR ESTACIÓN METEOROLÓGICA
- FRA.FV.MEM.COM.EM.007 MEMORIA DE DISEÑO ESTACIÓN METEOROLÓGICA

FORMATO							
A1	1A	09/02/2026	VCR	MLP	PJL	PSS	As Built
ESCALA	00	15/07/2025	VCR	MLP	PJL	PSS	Válido para construcción
S/E	0A	24/06/2025	VCR	MLP	PJL	PSS	Emisión Inicial
	Rev.	Fecha	Diseñado	Dibujado	Revisado	Aprobado	Descripción
		PLANTA FOTOVOLTAICA SOLAR FRAGA, HUESCA					CODIGO: FRA.FV.DRW.COM.EM.007 REVISION: 1A HOJA: 01
		DETALLE DE ESTACION METEOROLÓGICA					

GHI PYRANOMETER(S) -
INSTALLATION
SPECIFICATIONS

- Installed on horizontal boom at top of tower
- Pyranometer leveled within 0.5 degrees horizontal, using leveling feet
- Sun shield installed
- Sensor cable exit toward tower

TOWER DIMENSIONS
CENTRAL TOWER



Lookup ID	Description	Item Number	Qty. Complete Station	Qty Secondary Station
1	Lattice Tower, Lasser, 3m Flat top, C4	20219	1	1
2	NRG Pyranometer Pipe Mount Assembly	14357	3	3
3	RPS, One Panel, Lattice Tower	17029	1	1
4	Assembly, Lasser Lattice Tower, 4ft Boom	15946	1	1
5	Power Supply, Enclosure + LOGR Solar + Lattice Mounting Assembly	18324 + 9432 + 17170	1	1
6	Assembly, Lattice Tower Mounted Battery Bracket	18287	1	1
7	Sensor, Pyranometer, NRG R2-D, Digital	9462	5	5
8	Assembly, Pipe Mount, Pyranometer, Universal	14502	1	1
9	Delta-T Pyranometer, SPN1	10535	1	1
10	Assembly, RHI, Lattice Tower	16953	1	1
11	Assembly, Lattice Mounted Calypso	20136	1	1

Lookup ID	Description	Item Number	Qty. Complete Station	Qty Secondary Station
12	Thies Klima MultiMet	19501	1	1
13	PV Sensor, Reference Cell, IMT	15496	2	2
14	Sensor, Temperature, PV Panel, 50 m Cable	9424	6	5
15	Array Assembly, NRG SP100, Surge Device	20304	2	2
16	Rod-Ground, 5/8" x 4', Copper Clad Steel	1533	2	2
17	IMT Digital Back of Module Temp Sensor	19086	0	1
18	#40 Anemometer	14881	0	1
19	200M Wind Vane	14861	0	1
20	Assembly, POA Ref Cell and Pyranometer	17830	2	2
21	Side Mount Z Boom, Lattice, 0.9m	16937	0	2

DOCUMENTOS RELACIONADOS

- FRA.FV.DRW.COM.EM.006 UBICACIÓN DE ESTACIONES METEOROLÓGICAS
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- FRA.FV.MEM.COM.EM.007 MEMORIA DE DISEÑO ESTACIÓN METEOROLÓGICA

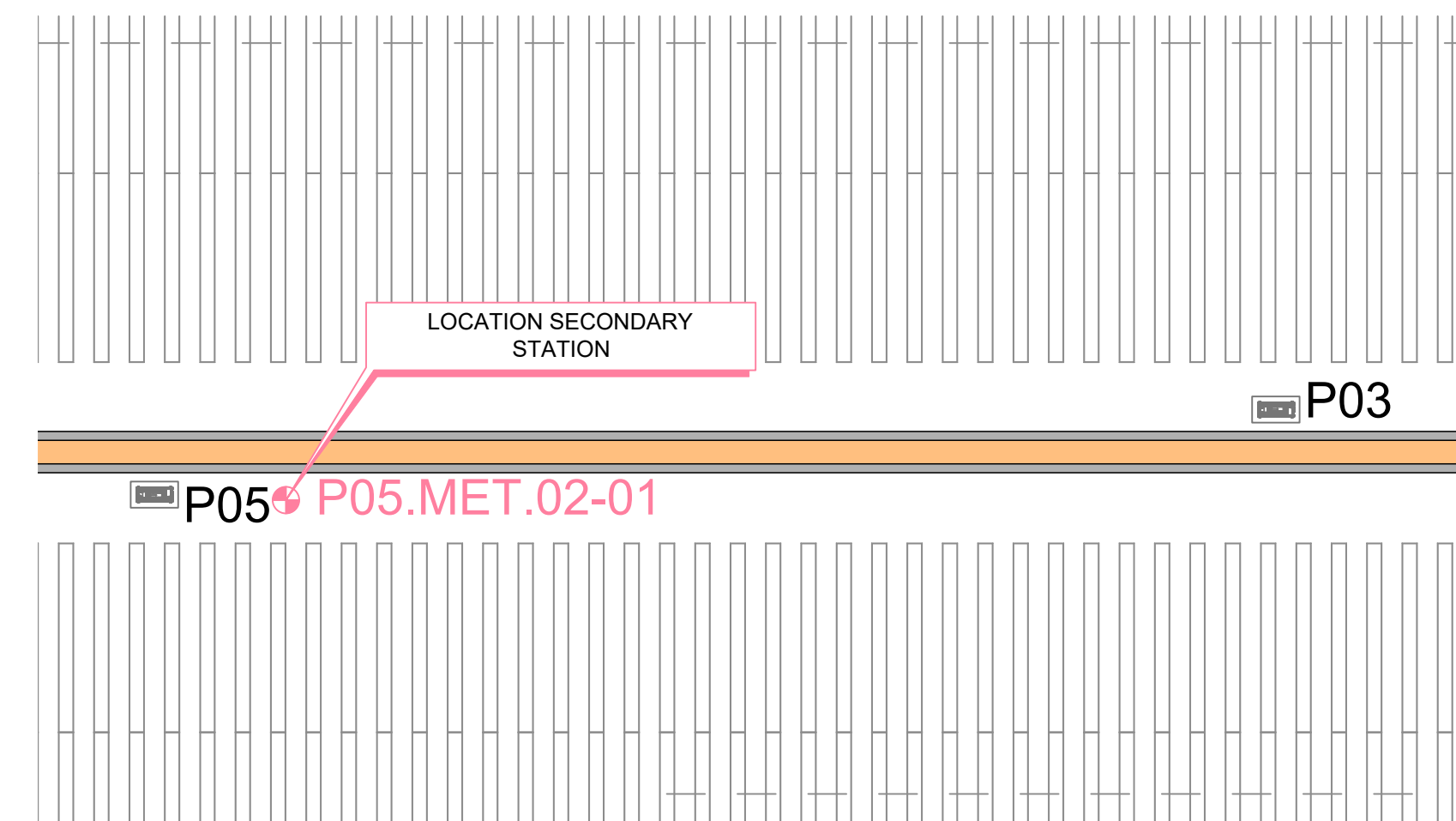
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ESCALA	00	15/07/2025	VCR	MLP	PJL	PSS	Válido para construcción
S/E	0A	24/06/2025	VCR	MLP	PJL	PSS	Emisión Inicial
	Rev.	Fecha	Diseñado	Dibujado	Revisado	Aprobado	Descripción
 		PLANTA FOTOVOLTAICA SOLAR FRAGA, HUESCA					CODIGO:
							FRA.FV.DRW.COM.EM.007
		DETALLE DE ESTACION METEOROLÓGICA					REVISION:
							1A
					HOJA:	02	

- Installed on horizontal boom at top of tower
- Pyranometer leveled within 0.5 degrees horizontal, using leveling feet
- Sun shield installed
- Sensor cable exit toward tower



PLAN VIEW
Units mm

This diagram shows the plan view of the mounting bracket assembly. It includes a large circle representing the field of view. A north arrow points to the left, labeled 'N'. Three callout numbers are present: 18 points to a screw on the left, 19 points to a screw on the top left, and 21 points to a screw on the top right. The assembly consists of a rectangular plate with various mounting points and a central circular feature.



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1	Lattice Tower, Lasser, 3m Flat top, C4	20219	1	1
2	NRG Pyranometer Pipe Mount Assembly	14357	3	3
3	RPS, One Panel, Lattice Tower	17029	1	1
4	Assembly, Lasser Lattice Tower, 4ft Boom	15946	1	1
5	Power Supply, Enclosure + LOGR Solar + Lattice Mounting Assembly	18324 + 9432 + 17170	1	1
6	Assembly, Lattice Tower Mounted Battery Bracket	18287	1	1
7	Sensor, Pyranometer, NRG R2-D, Digital	9462	5	5
8	Assembly, Pipe Mount, Pyranometer, Universal	14502	1	1
9	Delta-T Pyranometer, SPN1	10535	1	1
10	Assembly, RHI, Lattice Tower	16953	1	1
11	Assembly, Lattice Mounted Calypso	20136	1	1

Lookup ID	Description	Item Number	Qty. Complete Station	Qty Secondary Station
12	Thies Clima MultiMet	19501	1	1
13	PV Sensor, Reference Cell, IMT	15496	2	2
14	Sensor, Temperature, PV Panel, 50 m Cable	9424	6	5
15	Array Assembly, NRG SP100, Surge Device	20304	2	2
16	Rod-Ground, 5/8" x 4', Copper Clad Steel	1533	2	2
17	IMT Digital Back of Module Temp Sensor	19086	0	1
18	#40 Anemometer	14881	0	1
19	200M Wind Vane	14861	0	1
20	Assembly, POA Ref Cell and Pyranometer	17830	2	2
21	Side Mount Z Boom, Lattice, 0.9m	16937	0	2

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FORMATO							
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	Rev.	Fecha	Diseñado	Dibujado	Revisado	Aprobado	Descripción



PLANTA FOTOVOLTAICA SOLAR
FRAGA, HUESCA

DETALLE DE ESTACION METEOROLÓGICA

CODIGO:
FRA.FV.DRW.COM.EM.007

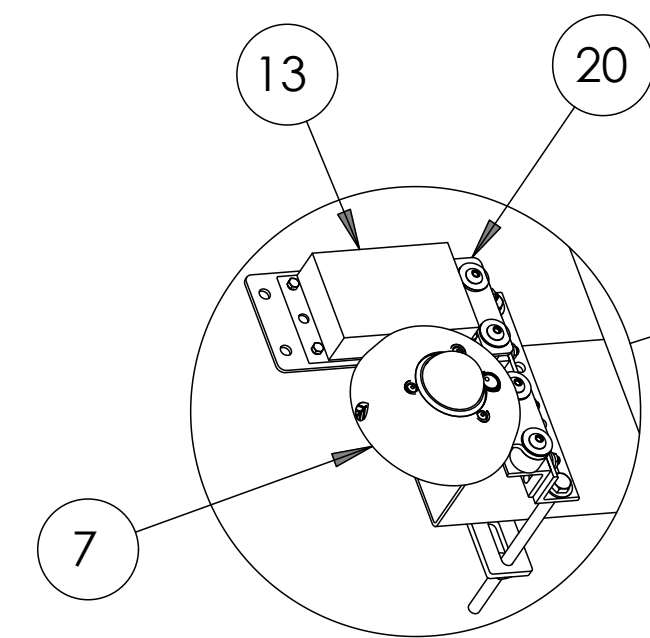
REVISION: 1A

HOJA: 03

ARRAY MOUNTED SENSOR DETAILS

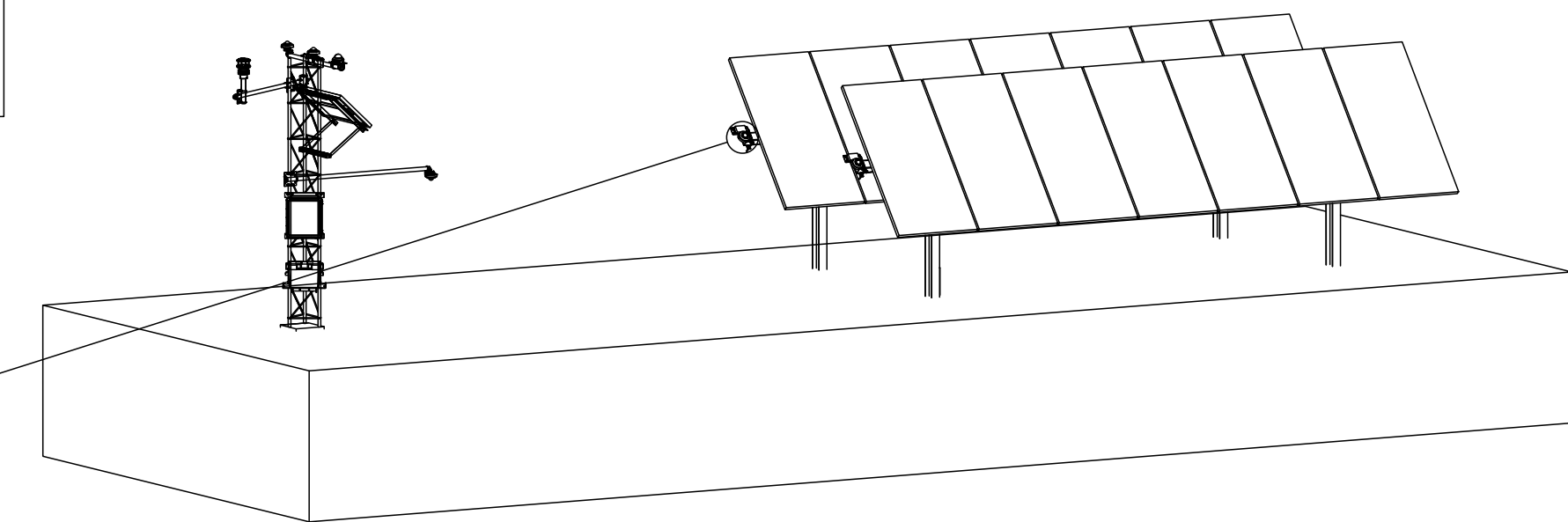
POA REFERENCE CELL(S) - INSTALLATION SPECIFICATIONS

- Installed on end of torque tube (requires > 12" of free space)
- Sensor cable exit toward nearest pole unless site constraints prohibit (otherwise install along tracking torque tube)
- 50m cable



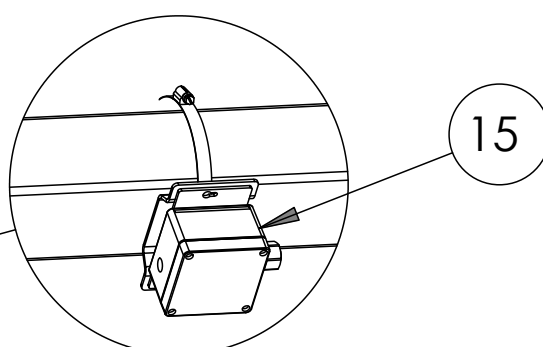
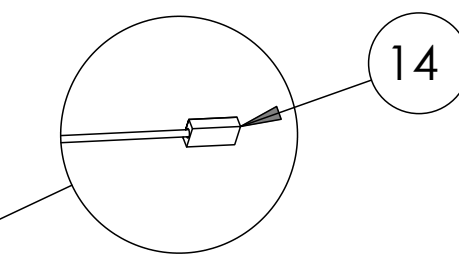
POA PYRANOMETER(S) - INSTALLATION SPECIFICATIONS

- Installed on end of torque tube (requires > 12" of free space)
- Remove mounting feet, level to array (within 0.5 degrees)
- Sensor cable exit toward nearest pole unless site constraints prohibit (otherwise install along tracking torque tube)
- Daisy chain sensor cables (if applicable), route cables through surge protection
- 50m cable



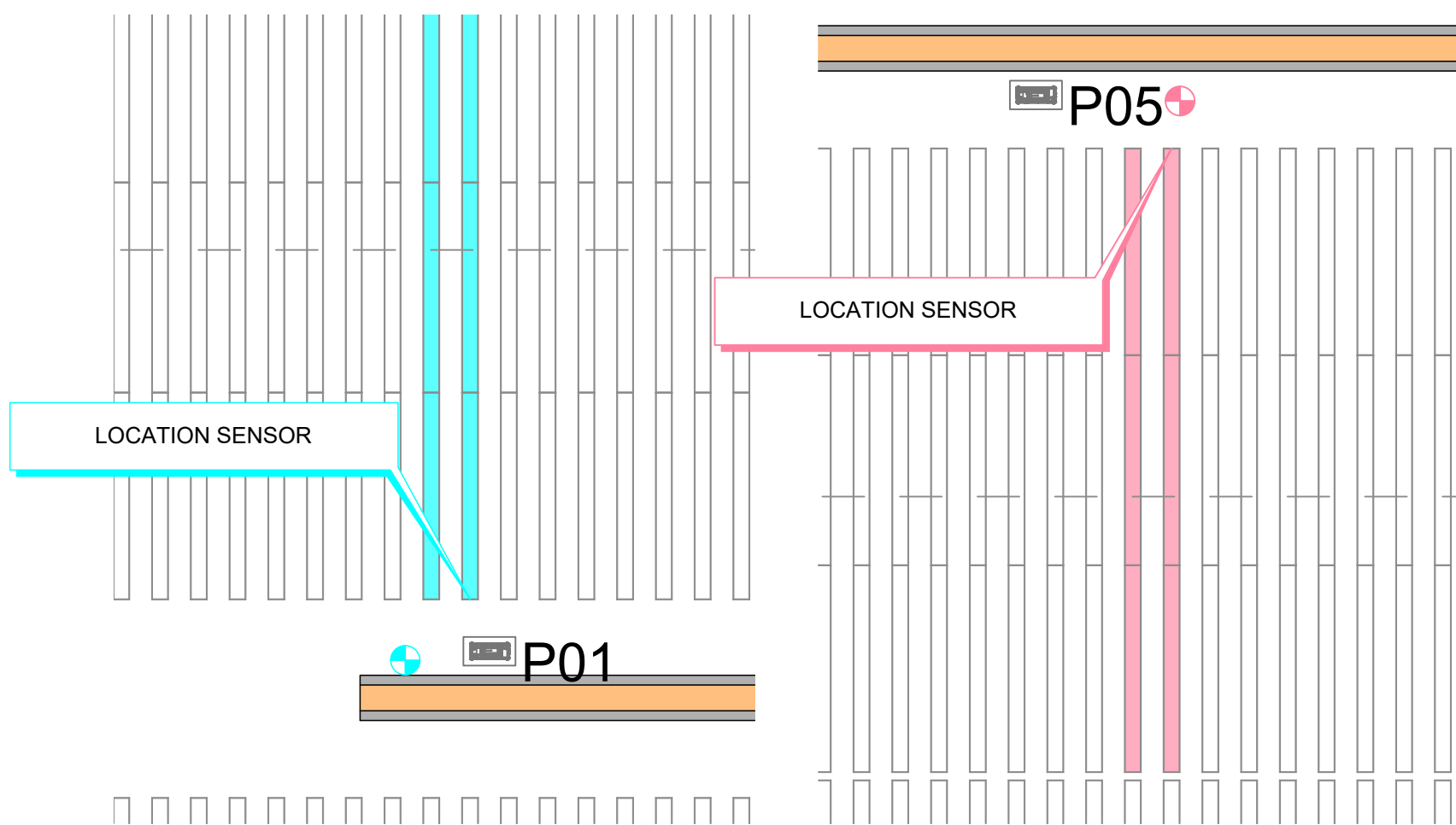
PVT1 BACK OF MODULE TEMP SENSORS - INSTALLATION SPECIFICATIONS

- Peel & adhere to back of panel
- Locate in center of center-most cell on panel
- Extra adhesive & strain relief provided
- EPC to determine optimal placement along string
- 50m cables



NRG SP100 SURGE DEVICE - INSTALLATION SPECIFICATIONS

- Hose clamp to torque tube
- Connect inline with array mounted pyranometer (device has 1m cable with connector)
- Must be grounded (6m grounding cable and 4ft ground rod provided, EPC to install ground rod)



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		DETALLE DE ESTACION METEOROLÓGICA					REVISION: 1A
							HOJA: 04